

Fingerprint Spoofing Detection

A Comparative Study of Generative and Discriminative Classifiers

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Abstract

This project addresses a binary classification problem: distinguishing genuine fingerprint images from spoofed (fake) ones, using a 6-dimensional feature representation produced by an image feature extractor. The dataset consists of 6000 labeled training samples (2990 fake, 3010 genuine) and a separate held-out evaluation set. A broad range of classifiers is developed and compared: generative Gaussian models (MVG, tied, naive Bayes), linear and quadratic logistic regression, linear and kernel support vector machines, and Gaussian mixture models, together with dimensionality reduction (PCA/LDA), decision-theoretic evaluation (minimum and actual DCF) under several operating points, score calibration, and model fusion. On the primary application ($\pi_T = 0.1$), the best single model is a full-covariance GMM (minDCF ≈ 0.150), while quadratic logistic regression and degree-4 polynomial / RBF kernel SVMs achieve comparable minDCF in the range 0.17–0.24. Calibration substantially improves the actual DCF of the SVM and logistic-regression scores, and the calibrated GMM is selected as the final delivered system, confirmed on the independent evaluation set.

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1 Introduction and Dataset

The task is fingerprint spoofing detection: given a 6-dimensional feature vector extracted from a fingerprint image, predict whether the fingerprint is *genuine* (class 1) or *counterfeit* (class 0). The training set contains 6000 samples (2990 labeled fake, 3010 labeled genuine); an independent evaluation set of 6000 samples is reserved for the final assessment of the delivered system and is never used for model selection or calibration.

For model development, the training data is split 2:1 into a *model training* portion (4000 samples) and a *validation* portion (2000 samples), using a fixed random seed so that the same split is reused across all classifiers, making results directly comparable.

2 Exploratory Feature Analysis

Figure 1 shows the normalized per-class histograms of all six features, and Figure 2 shows all fifteen pairwise scatter plots.

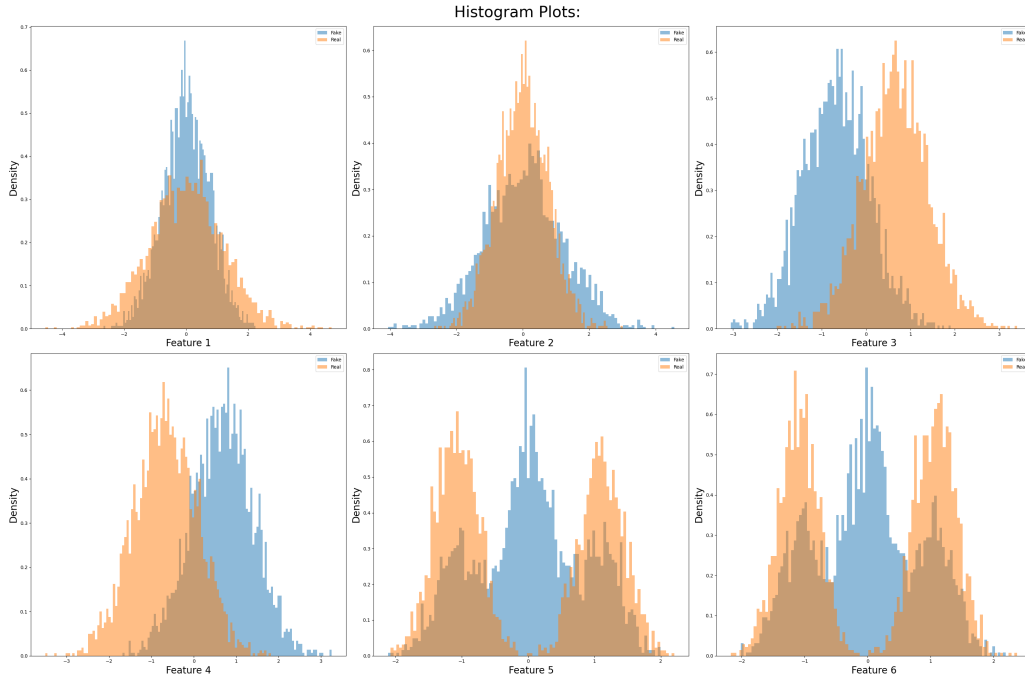


Figure 1: Normalized histograms of the six features, split by class (fake vs. real).

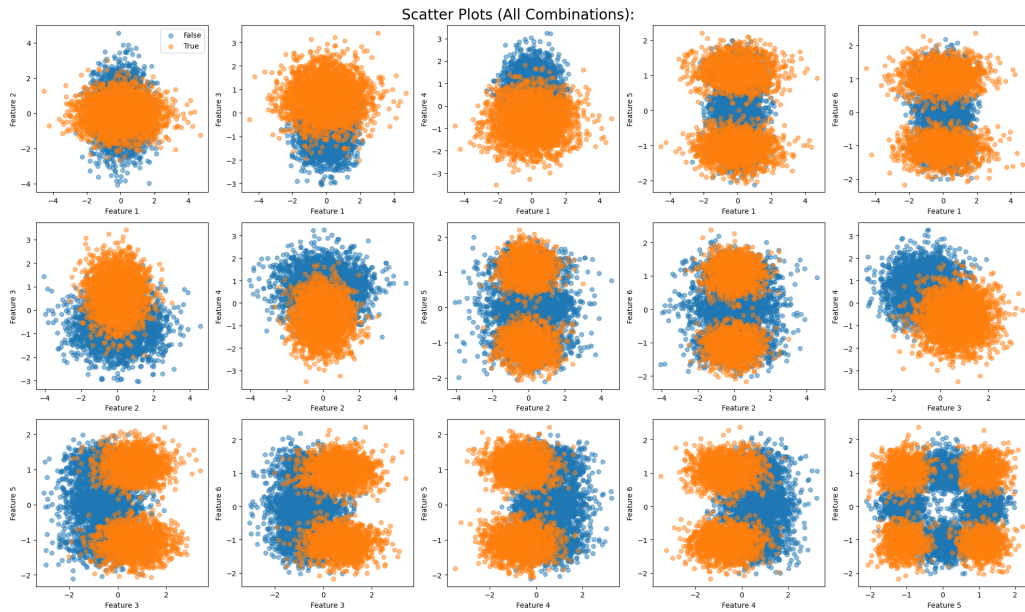


Figure 2: Pairwise scatter plots for all $\binom{6}{2} = 15$ feature combinations.

2.1 Features 1 and 2

The two classes overlap heavily for features 1 and 2, with both distributions concentrated near the center. The means are close for both features (class means differ by only 0.002 for feature 1 and 0.027 for feature 2, versus feature standard deviations around 0.75–1.2), so there is essentially no between-class shift to exploit. The variances, however, are *not* equal: for feature 1 the genuine class has variance 1.430 versus 0.570 for the fake class, and for feature 2 the fake class has variance 1.421 versus 0.578 for the genuine class – i.e., the two classes swap which one is "wide" and which is "narrow" between the two features. Both marginals are unimodal. Because the class-conditional Gaussians for these two features have almost identical means but different

variances, their densities cross at two points, producing three decision regions; however the "good" middle region is extremely narrow, so a linear classifier is expected to perform close to chance on these two features alone.

2.2 Features 3 and 4

Features 3 and 4 separate the classes much better. The class means are clearly shifted (feature 3: -0.681 for fake vs. $+0.665$ for genuine; feature 4: $+0.671$ for fake vs. -0.664 for genuine), while the variances are close to equal across classes (≈ 0.55 for both classes on both features). The overlap that remains is concentrated in the tails, where the two shifted, similarly-shaped bells meet. Both features are unimodal, and the equal-covariance/different-mean structure is exactly the situation a linear classifier (and in particular the tied-covariance Gaussian model) is designed for.

2.3 Features 5 and 6

Features 5 and 6 are qualitatively different: each per-class histogram is clearly multi-modal (roughly 2–3 peaks visible per class), and the scatter plot of feature 5 vs. feature 6 reveals a checkerboard-like pattern of (at least) four separable clusters, two per class, positioned so that the two classes' clusters interleave. Because a 1D histogram is a projection of the 2D cluster structure onto a single axis, two clusters that share the same horizontal position collapse into a single histogram peak – which is why the histograms show only 2–3 peaks even though the scatter plot reveals more distinct clusters. This distinction between what is visible in a 1D projection and what is present in the full joint distribution is an important limitation of purely marginal exploratory analysis: a linear classifier is expected to struggle with this multi-modal, multi-cluster structure, and single-Gaussian models are expected to fit these two features poorly (Section 4).

3 Dimensionality Reduction: PCA and LDA

3.1 Principal Component Analysis

PCA is applied to the full training set and the data is projected onto all six principal directions (Figure 3). The first direction shows the clearest class separation of all six PCA components, while the remaining directions show progressively more overlapping, less structured distributions.

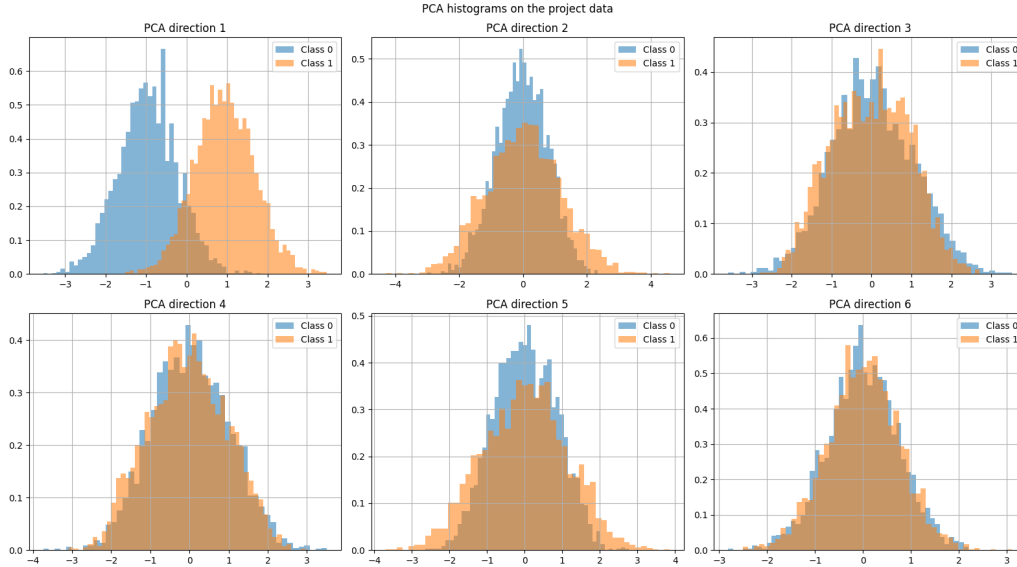


Figure 3: Histograms of the data projected onto each of the six PCA directions (ordered by decreasing variance), colored by class.

Crucially, PCA with all six components is only a rotation of the original feature space: it does not change the pairwise distances between samples, the cluster structure, or the true separability of the classes – only the angle from which that structure is viewed. The reason the first PCA direction happens to show good class separation on this dataset is a property of the data, not a general property of PCA: the within-class covariance matrices of both classes are approximately diagonal, with broadly similar values on the diagonal (see Section 5), which is precisely the special case in which the direction of maximum total variance ($S_T = S_B + S_W$, maximized by PCA) coincides with the direction of maximum linear discriminability (S_B/S_W , maximized by LDA). This is a coincidence of this particular dataset’s geometry rather than a guarantee that PCA finds discriminative directions in general.

3.2 Linear Discriminant Analysis

Because there are only two classes, LDA yields a single one-dimensional discriminant direction. Figure 4 shows the histogram of the training data projected onto this direction, together with the decision threshold placed at the midpoint of the two projected class means ($\text{thr} = -0.0104$). The class overlap along this direction is visibly smaller than for any of the six original features or any single PCA direction. This is expected: LDA explicitly uses the class labels to search for the direction that maximizes between-class separation relative to within-class spread, whereas PCA and the raw features do not target separability directly. Because features 1–2 have almost identical means for the two classes and features 5–6 have a poor linear structure, LDA places most of its discriminative weight on features 3 and 4, whose means are well separated and whose variances are similar across classes.

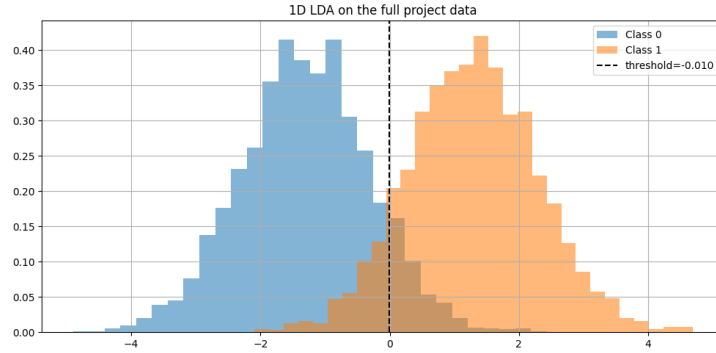


Figure 4: Histogram of the training data projected onto the 1D LDA direction, with the classification threshold (midpoint of the two projected class means) shown as a dashed line.

3.3 LDA as a Classifier

Using the midpoint threshold, LDA trained on the model-training split and evaluated on the validation split achieves an error rate of **9.30%** (186 errors out of 2000 validation samples). Sweeping the threshold over the range of projected validation scores gives a best achievable error rate of **9.15%** at threshold -0.109 (versus the default midpoint threshold of -0.0185), a marginal improvement, confirming that the default midpoint threshold is already close to optimal for this direction.

3.4 PCA as Pre-processing for LDA

Table 1 reports the validation error of PCA+LDA for PCA dimensionality $m \in \{2, 3, 4, 5, 6\}$ (PCA estimated on the training split only). Error rates are essentially flat around 9.25–9.30% regardless of m , confirming that discarding PCA directions neither helps nor substantially hurts LDA on this dataset – the discriminative direction that LDA needs is preserved across all tested values of m .

PCA dimensions m	Validation error
2	9.25%
3	9.25%
4	9.25%
5	9.30%
6	9.30%

Table 1: Validation error of PCA (estimated on the training split) followed by LDA classification, for varying number of retained PCA dimensions m .

4 Univariate Gaussian Density Fitting

Maximum-likelihood univariate Gaussian densities are fitted independently to each feature, for each class, using the full training set. Table 2 reports the estimated means and variances, and Figure 5 overlays the fitted densities on the normalized histograms for both classes simultaneously.

Class	Feature	Mean μ	Variance	Std. dev.
0 (fake)	1	0.0029	0.5696	0.7547
0 (fake)	2	0.0187	1.4209	1.1920
0 (fake)	3	-0.6809	0.5500	0.7416
0 (fake)	4	0.6708	0.5360	0.7321
0 (fake)	5	0.0280	0.6801	0.8247
0 (fake)	6	-0.0058	0.7050	0.8397
1 (genuine)	1	0.0005	1.4302	1.1959
1 (genuine)	2	-0.0085	0.5783	0.7604
1 (genuine)	3	0.6652	0.5489	0.7409
1 (genuine)	4	-0.6642	0.5533	0.7439
1 (genuine)	5	-0.0417	1.3178	1.1479
1 (genuine)	6	0.0239	1.2870	1.1345

Table 2: Maximum-likelihood univariate Gaussian parameters per class, per feature, estimated on the full training set.

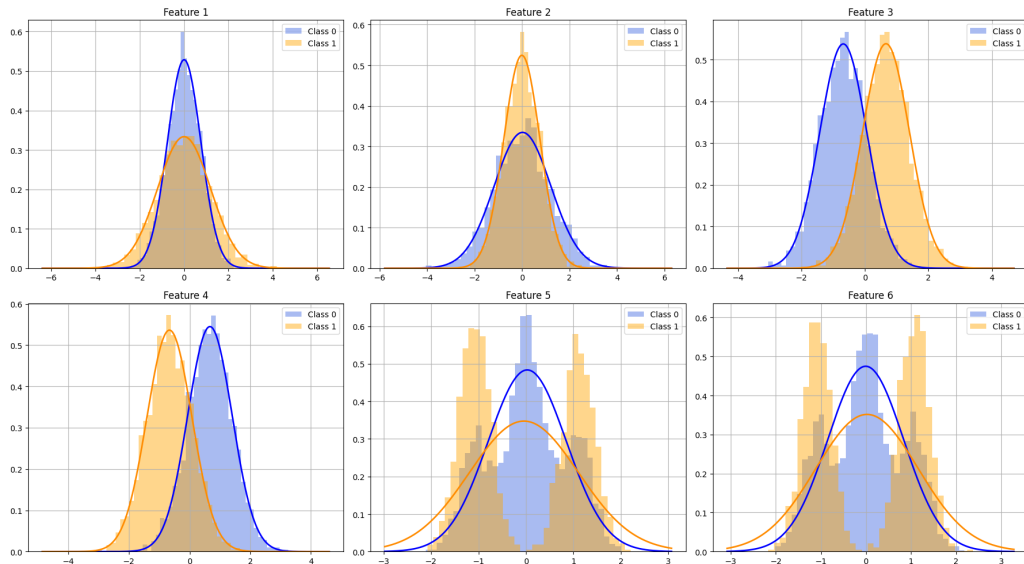


Figure 5: Fitted univariate Gaussian densities (curves) overlaid on the normalized per-class histograms, for all six features. Blue = fake, orange = genuine.

For features 1–4, the Gaussian densities track the histograms reasonably well: the shapes are approximately bell-curved and the fitted mean/variance capture the visible spread. For features 5 and 6, the fit is visibly poor: the true class-conditional distributions have two to three modes each (consistent with the 4-cluster structure seen in the scatter plots), so the single fitted Gaussian is forced to sit between the modes, matching neither peak well and placing excess probability mass in the low-density valley between them.

Where two fitted class-conditional Gaussians differ in *both* mean and variance (as for features 5 and 6), their densities cross at two points, producing three decision regions instead of one; even though the Gaussian shape itself is a poor description of the true multi-modal density, the resulting log-likelihood ratio can still point in roughly the right direction across most of the range where one class visibly dominates the other. This explains why, despite the inaccurate distributional assumption, models built on features 5 and 6 still contribute useful discriminative information rather than simply adding noise – a result confirmed quantitatively in Section 5.

5 Generative Gaussian Classifiers

Three generative Gaussian classifiers – full-covariance MVG, tied-covariance MVG, and diagonal-covariance Naive Bayes – are trained on the model-training split (4000 samples: 1998 fake, 2002 genuine) and evaluated on the validation split (2000 samples: 992 fake, 1008 genuine) using log-likelihood ratios thresholded at 0 (uniform priors, $C_{fn} = C_{fp} = 1$).

5.1 Full 6-Feature Models

Table 3 reports the validation error rates. MVG achieves the lowest error (7.00%), closely followed by Naive Bayes (7.20%); the tied model is clearly worse (9.30%), matching the LDA error rate almost exactly (9.20%–9.30%), since a tied-covariance Gaussian classifier and LDA define the same linear decision boundary and differ only in how the threshold along that boundary is chosen (theoretical threshold at LLR = 0 for tied Gaussian vs. the empirical midpoint of projected class means for LDA).

Model	Validation error rate
MVG	7.00%
Naive Bayes Gaussian	7.20%
LDA	9.20%
Tied Gaussian	9.30%

Table 3: Validation error rate for the three Gaussian classifiers and LDA, full 6-feature space, threshold at LLR = 0.

5.2 Covariance and Correlation Analysis

The class covariance matrices estimated by MVG (scaled $\times 100$ and rounded for readability) are:

$$100 C_0 \approx \begin{pmatrix} 60 & 0 & 1 & 1 & 1 & 1 \\ 0 & 144 & 1 & 1 & 2 & 2 \\ 1 & 1 & 56 & 0 & 0 & 1 \\ 1 & 1 & 0 & 54 & 0 & 1 \\ 1 & 2 & 0 & 0 & 69 & 1 \\ 1 & 2 & 1 & 1 & 1 & 68 \end{pmatrix}, \quad 100 C_1 \approx \begin{pmatrix} 144 & 1 & 0 & 1 & 1 & 0 \\ 1 & 55 & 1 & 0 & 1 & 1 \\ 0 & 1 & 55 & 2 & 0 & 1 \\ 1 & 0 & 2 & 56 & 1 & 3 \\ 1 & 1 & 0 & 1 & 134 & 1 \\ 0 & 1 & 1 & 3 & 1 & 130 \end{pmatrix}$$

In both classes the off-diagonal covariance terms are one to two orders of magnitude smaller than the diagonal variances, and the corresponding Pearson correlation matrices confirm this: the largest absolute off-diagonal correlation across either class is only about 0.049 (class 1, features 4–6), with most entries below 0.02. The features are therefore weakly to moderately correlated but far from strongly correlated – the within-class covariance is approximately diagonal. This directly explains why Naive Bayes (7.20%) performs almost identically to MVG (7.00%): since MVG already estimates a near-diagonal covariance, forcing it to be exactly diagonal changes little, and on this finite sample Naive Bayes can even be marginally more accurate by not spending capacity estimating the small off-diagonal terms, which are partly finite-sample noise around a true value of (or close to) zero.

5.3 Effect of Discarding Features 5–6

Table 4 repeats the three-model comparison using only features 1–4. All three error rates increase relative to using all six features (MVG: 7.00% $\rightarrow$ 7.95%; Tied: 9.30% $\rightarrow$ 9.50%; Naive

Bayes: 7.20% $\rightarrow$ 7.65%), showing that even though the Gaussian assumption is a poor description of features 5–6 individually, those two features still carry discriminative information that a Gaussian-based classifier can exploit – removing them makes every model worse, not better.

Model	Validation error (features 1–4 only)
MVG	7.95%
Naive Bayes Gaussian	7.65%
Tied Gaussian	9.50%

Table 4: Validation error using only features 1–4 (discarding features 5–6).

5.4 Features 1–2 versus Features 3–4

Table 5 compares MVG and Tied Gaussian trained on features 1–2 only and on features 3–4 only. For features 1–2 (near-identical means, different variances) MVG is far better than Tied (36.50% vs. 49.45% error) because MVG can exploit the variance difference between classes, which the tied model – by construction – discards. For features 3–4 (well-separated means, similar variances) the two models perform almost identically (9.45% vs. 9.40%), since the equal-covariance assumption underlying the tied model is approximately satisfied and the discriminative signal (the mean shift) is fully captured by a shared-covariance linear boundary. This is a direct, quantitative confirmation of the qualitative feature analysis of Section 2.

Feature subset	MVG error	Tied Gaussian error
Features 1–2	36.50%	49.45%
Features 3–4	9.45%	9.40%

Table 5: Validation error of MVG and Tied Gaussian restricted to features 1–2 and to features 3–4.

5.5 PCA as Pre-processing for the Gaussian Models

Table 6 sweeps PCA dimensionality m from 1 to 6 (PCA estimated on the training split) before each of the three Gaussian classifiers. The best configuration overall remains MVG with the full 6-dimensional feature space (7.00% error); PCA pre-processing does not improve on this. Naive Bayes is the model most negatively affected by PCA: its error rises from 7.20% (no PCA) to 8.90% at $m = 6$. This is because PCA diagonalizes the *total* covariance matrix (computed across both classes together), not the *within-class* covariance matrices that Naive Bayes needs to be diagonal; the two matrices are generally not diagonalized by the same rotation, so a rotation that decorrelates the pooled data can introduce new within-class correlations that were not present in the original feature space, directly violating the independence assumption Naive Bayes relies on.

PCA dim. m	MVG	Naive Bayes Gaussian	Tied Gaussian
No PCA (6)	7.00%	7.20%	9.30%
1	9.25%	9.25%	9.35%
2	8.80%	8.85%	9.25%
3	8.80%	9.00%	9.25%
4	8.05%	8.85%	9.25%
5	7.10%	8.75%	9.30%
6	7.00%	8.90%	9.30%

Table 6: Validation error of MVG, Naive Bayes, and Tied Gaussian as a function of PCA pre-processing dimensionality m , compared to the no-PCA baseline.

5.6 Summary

Across all Gaussian-model experiments, **MVG on the full 6-dimensional feature space** gives the best validation error (7.00%). The consistent gap between MVG/Naive Bayes and the Tied model reflects the fact that the two classes genuinely differ in covariance structure (most visibly for features 1–2 and 5–6), which the tied assumption cannot represent.

6 Bayes Decisions and Model Evaluation

6.1 Applications and Effective Priors

Table 7 reports the effective prior $\tilde{\pi} = \pi_1 C_{fn} / (\pi_1 C_{fn} + (1 - \pi_1) C_{fp})$ for the five specified applications. The five applications collapse onto only three distinct effective priors: $\tilde{\pi} \in \{0.1, 0.5, 0.9\}$. In particular, the "strong security" application (0.5, 1.0, 9.0) is equivalent to a uniform-cost application with effective prior 0.1 (a system biased toward rejecting, as if genuine users were rare), while the "ease of use" application (0.5, 9.0, 1.0) is equivalent to effective prior 0.9 (biased toward accepting).

(π_1, C_{fn}, C_{fp})	Description	Effective prior $\tilde{\pi}$
(0.5, 1.0, 1.0)	Uniform prior & costs	0.5000
(0.9, 1.0, 1.0)	High genuine prior	0.9000
(0.1, 1.0, 1.0)	High fake prior	0.1000
(0.5, 1.0, 9.0)	Strong security (high C_{fp})	0.1000
(0.5, 9.0, 1.0)	Ease of use (high C_{fn})	0.9000

Table 7: The five specified applications and their equivalent effective prior.

6.2 minDCF and actDCF at the Three Effective Priors

Table 8 reports minimum and actual DCF for MVG, Tied Gaussian, and Naive Bayes at the three effective priors, without any pre-processing.

Model	$\tilde{\pi} = 0.1$		$\tilde{\pi} = 0.5$		$\tilde{\pi} = 0.9$	
	actDCF	minDCF	actDCF	minDCF	actDCF	minDCF
MVG	0.305	0.263	0.140	0.130	0.400	0.342
Naive Bayes	0.302	0.257	0.144	0.131	0.389	0.351
Tied Gaussian	0.406	0.363	0.186	0.181	0.463	0.442

Table 8: Actual and minimum DCF for the three Gaussian models at the three effective priors, validation set, no pre-processing.

MVG and Naive Bayes clearly and consistently outperform the Tied model at all three operating points, in minDCF as well as actDCF, matching the error-rate ranking of Section 5. Calibration is reasonably good near the balanced application ($\tilde{\pi} = 0.5$): the actDCF–minDCF gap is under 0.015 for all three models there. The gap widens at the extreme priors ($\tilde{\pi} = 0.1$ and 0.9), especially for the Tied model (gap of 0.043 at $\tilde{\pi} = 0.1$ and 0.021 at $\tilde{\pi} = 0.9$), indicating that the theoretical threshold becomes progressively less well matched to the empirical score distribution away from the balanced operating point.

6.3 Bayes Error Plots

Figure 6 shows minimum and actual DCF for the three Gaussian models over the full range of prior log-odds $\in [-4, 4]$.

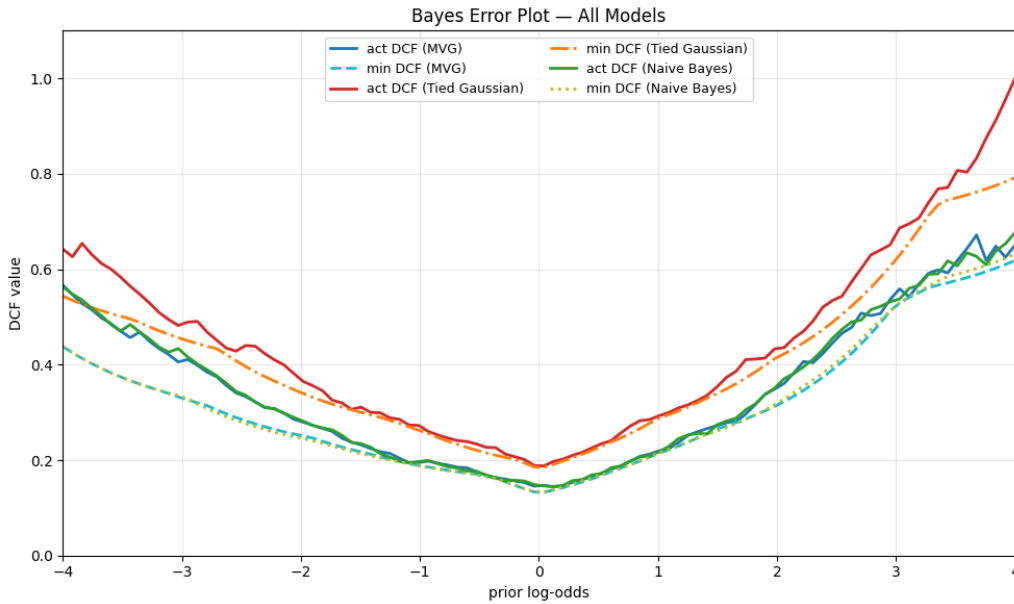


Figure 6: Bayes error plot: minimum (dashed) and actual (solid) DCF for MVG, Tied Gaussian, and Naive Bayes, as a function of prior log-odds.

The ranking by minimum DCF (MVG $\approx$ Naive Bayes $>$ Tied) is preserved across essentially the entire range of operating points, not just at the three sampled applications. All three models are best calibrated near the balanced operating point ($\tilde{p} = 0$) and show increasing actual-vs-minimum DCF divergence toward the extremes, with the Tied model showing the largest calibration loss throughout.

7 Logistic Regression

All logistic regression experiments target the primary application $\pi_T = 0.1$. The training split has 4000 samples with empirical prior $\pi_{emp} \approx 0.5005$.

7.1 Standard Logistic Regression (Full Dataset)

Figure 7 sweeps $\lambda \in \text{logspace}(-4, 2, 13)$. Minimum DCF is essentially flat around 0.361–0.365 across the entire range of λ : with 4000 training samples, regularization has almost no effect on the discriminative quality of the model. Actual DCF, in contrast, degrades sharply as λ increases, from about 0.40 at $\lambda = 10^{-4}$ to 1.00 for $\lambda \gtrsim 1$. This is a pure calibration effect: strong regularization shrinks the weight vector toward zero, compressing the score range so much that the theoretical threshold $t = -\log(\pi_T/(1 - \pi_T)) \approx 2.20$ falls outside the range of achievable scores, causing all validation samples to be assigned to one class.

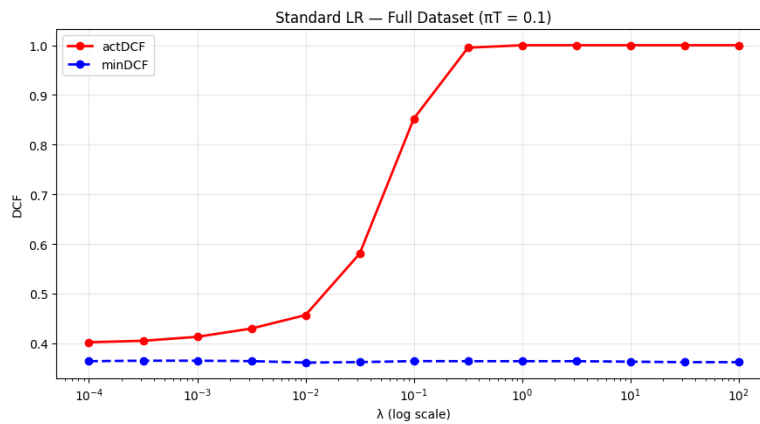


Figure 7: Standard logistic regression, full dataset: actDCF and minDCF vs. λ (primary application $\pi_T = 0.1$).

7.2 Effect of a Reduced Training Set

Repeating the sweep after subsampling the training split to 1 sample in 50 (80 training samples; Figure 8) makes the role of λ far more visible: at very low λ the model overfits (actDCF ≈ 0.98 , close to the trivial baseline), performance improves to a minimum around $\lambda \approx 10^{-2}$ (actDCF ≈ 0.47 , minDCF ≈ 0.44), and then degrades again as λ grows large enough to underfit. This confirms that the usefulness of regularization depends on the ratio between model complexity and the amount of available training data, not on model complexity alone.

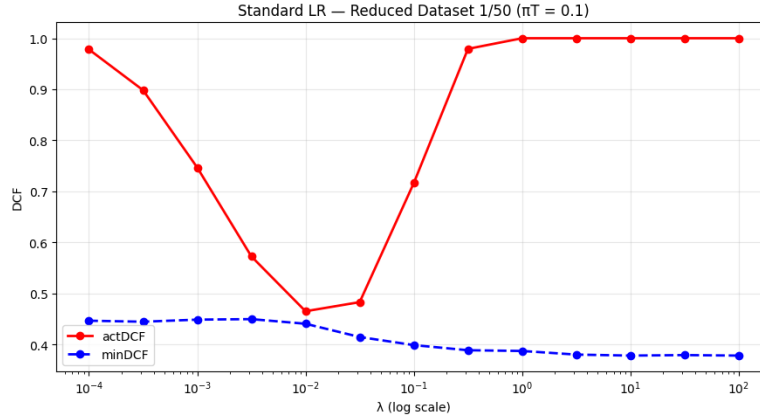


Figure 8: Standard logistic regression, reduced training set (1 sample in 50, 80 samples): actDCF and minDCF vs. λ .

7.3 Prior-Weighted Logistic Regression

Figure 9 shows the same sweep for the prior-weighted formulation, trained directly with $\pi_T = 0.1$. The curves are nearly indistinguishable from the standard (unweighted) model – best minDCF around 0.363 vs. 0.361 for standard LR. Because the training set is large and balanced ($\pi_{emp} \approx 0.5$), the linear boundary is driven mainly by feature geometry rather than by class imbalance, so re-weighting toward the target prior changes the learned boundary very little on this dataset.

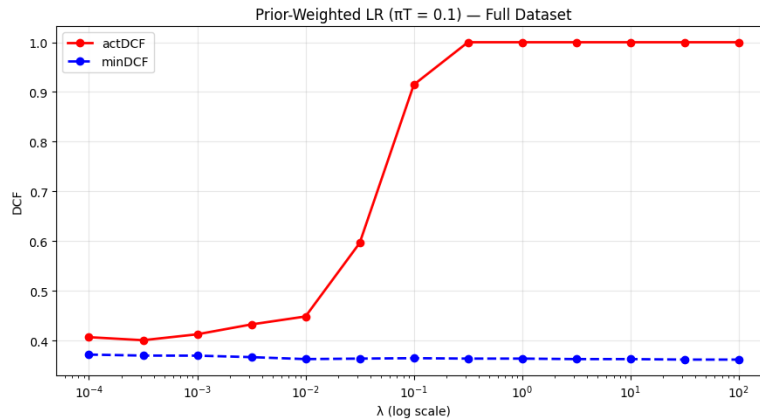


Figure 9: Prior-weighted logistic regression ($\pi_T = 0.1$), full dataset: actDCF and minDCF vs. λ .

7.4 Quadratic Logistic Regression

Expanding each 6-dimensional feature vector to the 42-dimensional quadratic feature map $\phi(\mathbf{x}) = [\text{vec}(\mathbf{x}\mathbf{x}^\top), \mathbf{x}]$ and retraining standard logistic regression (Figure 10) gives a markedly different picture: minDCF now shows a clear minimum of **0.244** around $\lambda \approx 3 \times 10^{-2}$, degrading toward 0.326 for large λ and toward 0.260 (with poor calibration) for very small λ . Unlike the linear model, regularization is genuinely useful here, because the 42-dimensional expanded space has enough capacity to overfit the 4000 training samples. Quadratic terms $x_i x_j$ let the model represent elliptical/curved decision boundaries – effectively the same class of boundary that MVG’s covariance matrices provide – which is why quadratic LR closes most of the gap to (and ultimately exceeds) the Gaussian models on this feature set, where the discriminative signal between classes in features 1–2 and 5–6 is carried by variance and multi-modal structure rather than by mean shift alone.

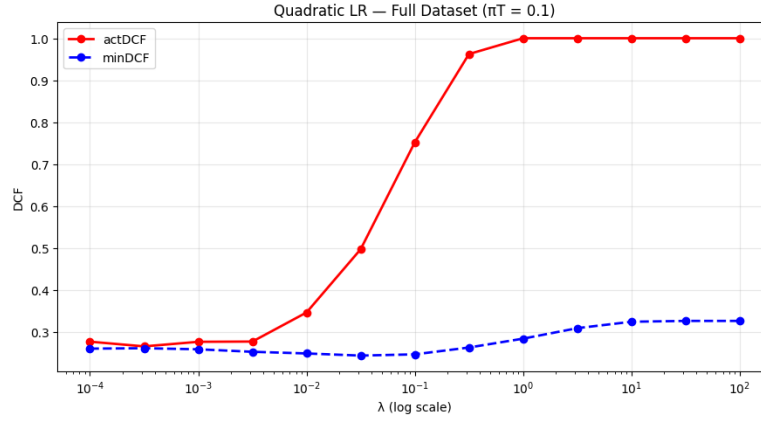


Figure 10: Quadratic logistic regression, full dataset: actDCF and minDCF vs. λ .

7.5 Comparison of All Models So Far

Figure 11 and Table 9 compare the best- λ minDCF of every logistic-regression variant against the three Gaussian classifiers from Section 5, all at the primary application $\pi_T = 0.1$.

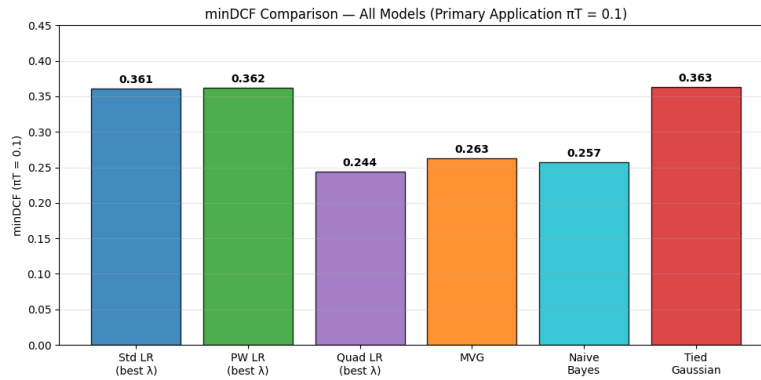


Figure 11: Best- λ minDCF comparison across logistic-regression variants and Gaussian models ($\pi_T = 0.1$).

Model	Best minDCF ($\pi_T = 0.1$)
Quadratic Logistic Regression	0.244
Naive Bayes Gaussian	0.257
MVG	0.263
Standard Logistic Regression	0.361
Prior-Weighted Logistic Regression	0.362
Tied Gaussian	0.363

Table 9: Best- λ minDCF for every model trained up to this point, primary application $\pi_T = 0.1$.

Quadratic logistic regression achieves the best minDCF of all models considered so far. This is fully consistent with the earlier feature analysis: linear models and the tied Gaussian classifier are limited to ≈ 0.36 minDCF because they cannot represent the variance-driven separability of features 1–2 or the multi-modal structure of features 5–6, whereas quadratic LR – like MVG’s full covariance matrices – can.

8 Support Vector Machines

All SVM experiments target the primary application $\pi_T = 0.1$.

8.1 Linear SVM ($K = 1$)

Figure 12 sweeps $C \in \text{logspace}(-5, 0, 11)$ for the linear SVM. Minimum DCF drops sharply once C is large enough to escape the trivial-classifier regime and then plateaus around 0.358–0.365 for $C \gtrsim 3 \times 10^{-5}$; the best value obtained is **0.358** at $C = 1.0$. Actual DCF, however, stays poor (close to 1.0) for small-to-moderate C and only descends to ≈ 0.49 at the largest tested C – SVM scores have no probabilistic interpretation by construction, so a large actDCF–minDCF gap is expected unless the scores are separately calibrated.

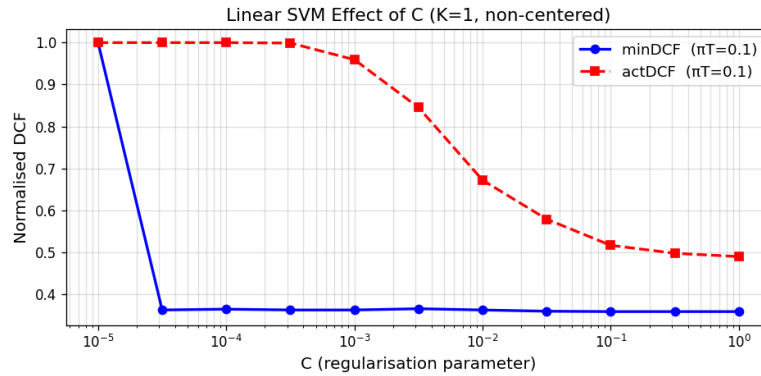


Figure 12: Linear SVM ($K = 1$, non-centered features): actDCF and minDCF vs. C .

Repeating the sweep on centered data (subtracting the training-set mean) gives essentially identical results (best minDCF 0.358 at the same C), confirming that centering has negligible effect once the bias term is included in the SVM formulation. The best linear-SVM minDCF (0.358) is close to, but slightly better than, the best linear logistic regression result (0.361), consistent with both being linear classifiers operating on the same feature space.

8.2 Polynomial Kernel SVM (degree 2)

With the degree-2 polynomial kernel ($d = 2, c = 1, \xi = 0$; Figure 13), minDCF improves substantially to a best value of **0.245** (at $C \approx 3 \times 10^{-2}$ and again at $C \approx 3 \times 10^{-5}$), matching the quadratic logistic regression result almost exactly – as expected, since both models are effectively fitting quadratic decision boundaries in the original feature space. Actual DCF remains poorly calibrated across the sweep (falling only to ≈ 0.39 at the largest C), reinforcing that kernel SVM scores require calibration before being used with a Bayes-optimal threshold.

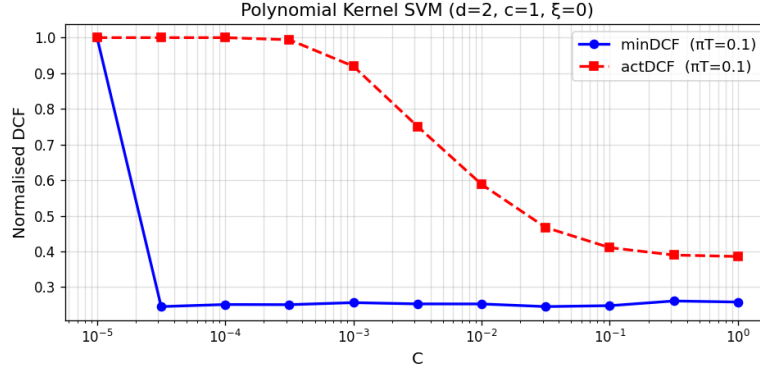


Figure 13: Polynomial kernel SVM ($d = 2, c = 1, \xi = 0$): actDCF and minDCF vs. C .

As an additional experiment, a degree-4 polynomial kernel was also trained. It further improves minDCF to **0.177** at $C \approx 3 \times 10^{-3}$, the best result obtained by any SVM variant, indicating that richer polynomial interactions continue to extract useful structure from this dataset (consistent with features 5–6 being separable by higher-order combinations of the raw features).

8.3 RBF Kernel SVM

Figure 14 shows the grid search over $\gamma \in \{e^{-4}, e^{-3}, e^{-2}, e^{-1}\}$ and $C \in \text{logspace}(-3, 2, 11)$ (with $\xi = 1$). The best region of the grid is around $\gamma = e^{-2}$ – e^{-1} and C in the range 1–30, where minDCF drops to approximately 0.18–0.19 – comparable to the degree-4 polynomial kernel and clearly better than the linear SVM or degree-2 polynomial kernel. Very small γ (broad, near-linear kernel) tracks the linear-SVM minDCF closely, while very large C combined with large γ starts to degrade minDCF again, consistent with overfitting from an overly localized kernel. As with the other SVM variants, actDCF remains substantially worse than minDCF across most of the grid, confirming that calibration is essential before deploying any of these kernel models directly.

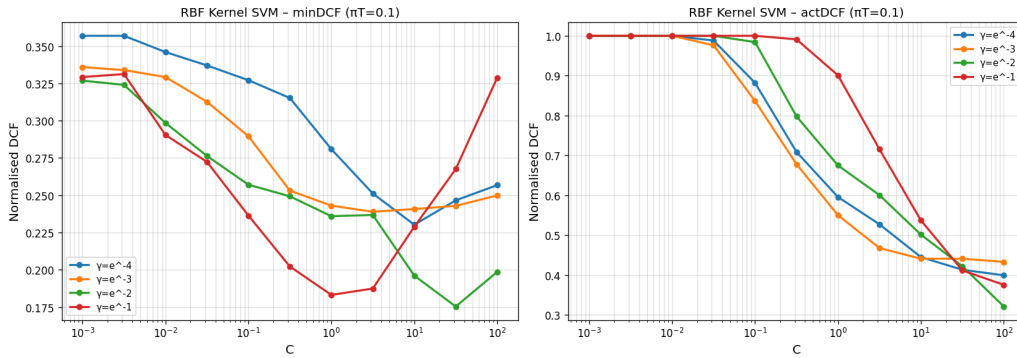


Figure 14: RBF kernel SVM grid search: minDCF (left) and actDCF (right) vs. C , one curve per γ .

8.4 Summary

Table 10 collects the best minDCF obtained by each SVM variant. Non-linear kernels (polynomial degree ≥ 2 , RBF) substantially outperform the linear SVM, mirroring the improvement seen from linear to quadratic logistic regression, and confirming that this dataset requires a non-linear decision boundary to be classified well.

SVM variant	Best minDCF ($\pi_T = 0.1$)
Linear (non-centered / centered)	0.358
Polynomial, $d = 2, c = 1$	0.245
RBF (best γ, C)	≈ 0.18
Polynomial, $d = 4, c = 1$	0.177

Table 10: Best minDCF achieved by each SVM configuration, primary application $\pi_T = 0.1$.

9 Gaussian Mixture Models

Full-covariance GMMs are trained independently per class using an LBG+EM procedure, sweeping the number of mixture components in $\{1, 2, 4, 8, 16\}$ for each class independently (25 combinations total), and selected by minimum DCF on the validation set at the primary application ($\pi_T = 0.1$). Table 11 reports the top results.

Rank	Components (class 0)	Components (class 1)	minDCF	actDCF
1	1	16	0.1495	0.2055
2	8	16	0.1527	0.1725
3	16	16	0.1631	0.1766
4	16	2	0.1657	0.1753
5	16	1	0.1670	0.1783

Table 11: Top 5 GMM configurations by validation minDCF (out of 25 combinations tested), primary application $\pi_T = 0.1$.

The best configuration uses a single component for the fake class and 16 components for the genuine class. More broadly, the best-performing combinations are consistently asymmetric, with more components assigned to whichever class benefits most from extra flexibility, and configurations with very few components on both classes (e.g., 1+1: minDCF 0.263; 1+2: minDCF 0.265) perform far worse – consistent with the earlier finding that a single Gaussian per class is a poor model given the multi-modal structure of features 5–6. This is also, by a clear margin, the best minDCF obtained by any model in this project. Larger mixtures do not uniformly help, however: several 16-component configurations show actDCF noticeably larger than minDCF (mild overfitting to the training distribution), and the very best minDCF configuration (1+16) is not the best-calibrated one (8+16 has a smaller actDCF-minDCF gap), illustrating that model selection by minDCF alone does not guarantee the best-calibrated model.

Figure 15 shows the Bayes error plot for the best GMM model, and Figure 16 compares it against the best logistic regression (quadratic, best λ) and best SVM (RBF grid search) models over the same wide range of operating points.

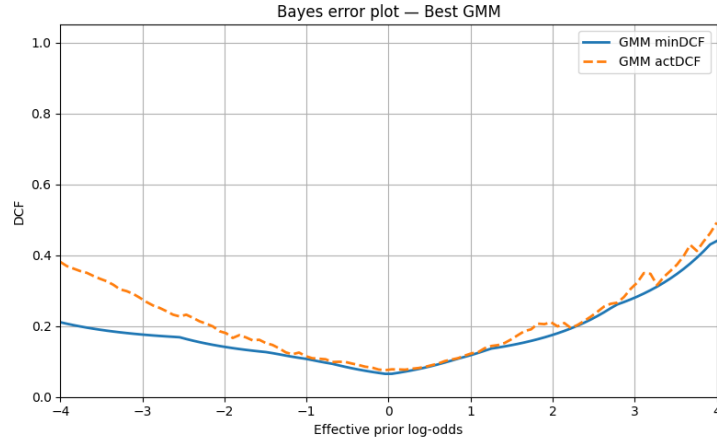


Figure 15: Bayes error plot for the best full-covariance GMM (class 0: 1 component, class 1: 16 components).

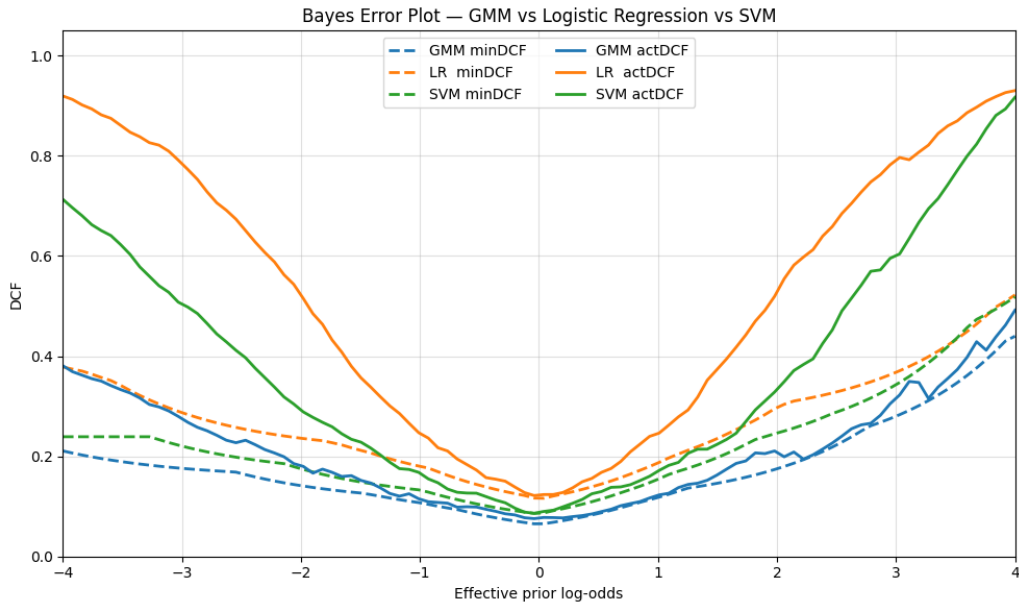


Figure 16: Bayes error plot comparing the best GMM, best (quadratic) logistic regression, and best (RBF) SVM models, over prior log-odds $\in [-4, 4]$.

The GMM’s minDCF curve sits below the logistic-regression and SVM curves across virtually the entire range of operating points, confirming that its advantage is not limited to the target application. In terms of actual DCF, the GMM is also the best-calibrated of the three families over most of the range, while the uncalibrated logistic-regression and SVM scores show a substantial and roughly symmetric gap between actual and minimum DCF at both extremes of the log-odds range – motivating the calibration stage of Section 11.

10 Model Comparison Across Approaches

Collecting the best model from each family evaluated at the primary application ($\pi_T = 0.1$):

Family	Best configuration	minDCF
GMM (full covariance)	1 comp. (class 0), 16 comp. (class 1)	0.150
SVM (polynomial, $d = 4$)	$C \approx 3 \times 10^{-3}$	0.177
SVM (RBF)	$\gamma \approx e^{-2}$, $C \approx 10\text{--}30$	≈ 0.18
Logistic Regression (quadratic)	$\lambda \approx 3 \times 10^{-2}$	0.244
Naive Bayes Gaussian	full 6 features	0.257
MVG	full 6 features	0.263

Table 12: Best minDCF achieved by each model family, primary application $\pi_T = 0.1$.

Non-linear/multi-component models (GMM, higher-degree kernels) clearly dominate purely linear models and single-Gaussian generative models on this task, in line with the exploratory finding that at least two of the six features (5 and 6) are only well described by multi-modal, cluster-structured distributions that linear or single-Gaussian assumptions cannot capture. The GMM emerges as the strongest single family, consistent with it being the most flexible density model among those tested.

11 Score Calibration and Fusion

For this stage the best model from each of the three families – logistic regression (quadratic, $\lambda = 10^{-2}$), linear SVM ($C = 10^{-1}$), and GMM (4 components per class) – is retrained, and its scores are calibrated with a K-fold (5-fold) prior-weighted logistic regression calibration transformation, trained with the target application prior $\pi_T = 0.2$ used for this stage of the project.¹

11.1 Raw and Calibrated Validation Scores

Table 13 reports minimum and actual DCF before and after calibration, on the pooled K-fold validation scores.

Model	minDCF (raw)	actDCF (raw)	minDCF (calibrated)	actDCF (calibrated)
Logistic Regression	0.2920	0.3084	0.2967	0.3073
SVM (linear)	0.2888	0.5596	0.2896	0.3073
GMM	0.1357	0.1377	0.1377	0.1417

Table 13: Minimum and actual DCF before and after K-fold calibration, $\pi_T = 0.2$, pooled validation scores.

Calibration is most impactful for the SVM: its raw actDCF (0.560) is far worse than its minDCF (0.289), indicating that the score ranking is good but the score scale is badly mismatched to the Bayes threshold; after calibration, actDCF drops to 0.307, essentially matching the calibrated logistic regression model and closing most of the calibration gap. Logistic regression changes little with calibration, since its scores were already reasonably close to LLR-like. The GMM was already well-calibrated before this step (actDCF 0.138 vs. minDCF 0.136), so the K-fold calibration transform adds only a small overhead (actDCF rises slightly to 0.142) rather than a benefit – consistent with GMM being a generative model that directly estimates class-conditional densities.

¹This calibration and fusion stage uses $\pi_T = 0.2$ as the target application prior, distinct from the $\pi_T = 0.1$ used for model selection in the earlier per-family comparisons.

11.2 Fusion

A score-level fusion is built by stacking the three (raw) scores and training a K-fold prior-weighted logistic regression on the resulting 3-dimensional score vectors, again at $\pi_T = 0.2$. On the pooled validation folds, the fusion achieves $\text{minDCF} = 0.1377$ and $\text{actDCF} = 0.1428$ – essentially matching, but not improving on, the calibrated GMM alone ($\text{minDCF} 0.1377$, $\text{actDCF} 0.1417$). The fusion is well calibrated but does not provide additional discriminative power beyond what the GMM already captures, suggesting that the three systems are not sufficiently complementary/diverse on this dataset for their combination to outperform the strongest individual model.

11.3 Delivered System

Based on *actual* DCF – the appropriate criterion once calibration has been addressed – the **calibrated GMM** is selected as the final delivered system. It achieves the best actual DCF of any individual model or the fusion on the validation folds, and (as shown below) this advantage is confirmed on the independent evaluation set.

12 Final Evaluation

The calibration transforms and the fusion model, trained exclusively on the training-set validation split, are applied unchanged to the independent 6000-sample evaluation set. Table 14 reports the results.

System	minDCF	actDCF
Logistic Regression (raw)	0.3899	0.3991
Logistic Regression (calibrated)	0.3899	0.4004
SVM (raw)	0.3876	0.5909
SVM (calibrated)	0.3876	0.4067
GMM (raw)	0.1536	0.1664
GMM (calibrated)	0.1536	0.1684
Fusion	0.1615	0.1734

Table 14: Minimum and actual DCF on the held-out evaluation set, $\pi_T = 0.2$.

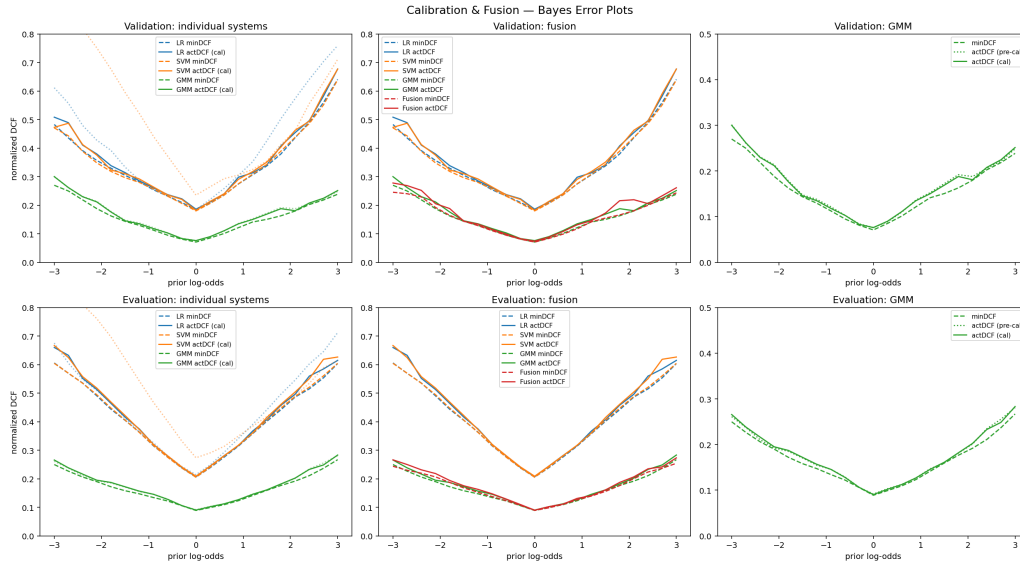


Figure 17: Bayes error plots comparing raw and calibrated scores for logistic regression, SVM, GMM, and their fusion, on the validation folds (top row) and on the evaluation set (bottom row).

12.1 Delivered System Performance

On the evaluation set, the GMM (calibrated) achieves $\text{minDCF} = 0.1536$ and $\text{actDCF} = 0.1684$ – a calibration loss of about 0.015, comparable to what was observed on the validation folds, indicating that the calibration transform generalizes well to unseen data. As shown in the bottom-right panel of Figure 17, the GMM’s actual DCF tracks its minimum DCF closely across nearly the entire range of prior log-odds on the evaluation set, confirming good calibration well beyond the single target application.

12.2 Comparison with the Fusion and Other Systems

The calibrated GMM alone ($\text{actDCF} = 0.1684$) outperforms the fusion ($\text{actDCF} = 0.1734$) on the evaluation set, mirroring the validation-fold result and confirming that the decision to deliver the GMM alone, rather than the fusion, was an effective one for this dataset: logistic regression and SVM contribute little complementary discriminative signal beyond what the GMM already captures, so including them in the fusion adds calibration risk without a compensating accuracy gain. The bottom-left and bottom-middle panels of Figure 17 show that both individual linear models (logistic regression, SVM) and their fusion with the GMM remain far worse than the GMM alone across the operating range, reinforcing this conclusion.

12.3 Calibration Effectiveness Across Systems

Calibration is most clearly beneficial for the SVM: its raw scores show severe miscalibration on the evaluation set ($\text{actDCF} = 0.591$ vs. $\text{minDCF} = 0.388$), while the calibrated scores are much closer to the minimum (0.407 vs. 0.388), and this improvement is visible across a wide range of prior log-odds in Figure 17 (top-left/bottom-left panels), not just at $\pi_T = 0.2$. Logistic regression was already close to calibrated ($\text{actDCF} = 0.399$ vs. $\text{minDCF} = 0.390$ raw), so calibration changes it only marginally. The GMM remains the best-calibrated system throughout, both before and after the explicit calibration step, consistent with its status as a generative model whose LLR scores already have a probabilistic interpretation by construction.

13 Conclusions

This project developed and systematically compared generative Gaussian models, linear and quadratic logistic regression, linear and kernel SVMs, and Gaussian mixture models for fingerprint spoofing detection, together with dimensionality reduction, multi-application decision-theoretic evaluation, score calibration, and model fusion.

The exploratory feature analysis proved highly predictive of downstream model performance throughout the project. The near-identical class means but different variances of features 1–2 explained why linear classifiers and the tied-covariance Gaussian model perform poorly on those two features in isolation (36.5%–49.5% error), while the well-separated means and matched variances of features 3–4 explained why linear methods work almost as well as full-covariance models there (9.40–9.45% error for both MVG and Tied). The clearly multi-modal, multi-cluster structure of features 5–6 explained both the poor univariate Gaussian fit for those features and the substantial performance gains obtained by every model family capable of representing non-linear or multi-component structure: quadratic logistic regression (minDCF 0.244 vs. 0.361 for the linear model), polynomial and RBF-kernel SVMs (minDCF as low as 0.177 vs. 0.358 for the linear SVM), and – most effectively of all – Gaussian mixture models (minDCF 0.150, the best result obtained in this project). The weak within-class feature correlation (largest Pearson correlation ≈ 0.05) explained why the Naive Bayes independence assumption is almost harmless on the raw features (7.20% vs. 7.00% error for MVG) yet becomes actively harmful after PCA rotation (error rising to 8.9% at full dimensionality), since PCA diagonalizes the pooled covariance rather than the within-class covariance that Naive Bayes actually requires.

Across all classifiers, the best minimum DCF at the primary application ($\pi_T = 0.1$) is achieved by the full-covariance GMM (0.150), ahead of degree-4 polynomial and RBF SVMs (≈ 0.18), quadratic logistic regression (0.244), and the single-Gaussian generative models (0.26–0.36). Calibration proved essential for the discriminative models (SVM in particular, whose raw actDCF of 0.56–0.59 fell to ≈ 0.31 –0.41 after calibration) but added little value for the already well-calibrated GMM. Fusing the three best-performing systems did not outperform the calibrated GMM alone, on either the validation folds or the independent evaluation set, indicating limited complementary information across the three model families for this task. The calibrated GMM was therefore selected as, and confirmed to be, the most effective delivered system, generalizing from the validation split to the held-out evaluation set with only a small, consistent calibration loss across a wide range of operating points.